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\documentclass[12pt]{article}

\usepackage{amsmath, amssymb, siunitx}

\usepackage[margin=1in]{geometry}

\title{NCHE 312 Physical Chemistry -- Class Test 5 \ Marking Memo}

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\date{22 May 2025}

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\begin{document}

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\maketitle

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\section*{Question 1 (5 marks)}

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Reaction: $2A + B \rightarrow 2C + 3D$ \

Given rate law:

$$\frac{d[C]}{dt} = k[A][B][C]$$

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\subsection*{(i) Reaction rate $v$}

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$$v = \frac{1}{\nu_C} \frac{d[C]}{dt} = \frac{1}{2} \frac{d[C]}{dt}$$

$$v = \frac{1}{2} k[A][B][C]$$

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\textbf{[3 marks]}

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\subsection*{(ii) Units of $k$}

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$\frac{d[C]}{dt}$ \text{ has units: } $\text{mol per cubic decimetre per second} = \text{molar per second}$

$[A][B][C]$ \text{ has units: } molar^3

$k = \frac{\text{molar per second}}{\text{molar}^3} = \text{molar}^{-2} \text{ per second}$

In $\text{dm}^3 \text{ per mol}$ units:

$\boxed{\text{dm}^6 \text{ per mol}^2 \text{ per second}}$

[2 marks]

Question 2 (5 marks)

Reaction: $A + 3B \rightarrow C + 2D$

Given:

$-\frac{d[B]}{dt} = 2.7 \text{ mol per dm}^3 \text{ per second}$

(i) Reaction rate

$v = -\frac{1}{3} \frac{d[B]}{dt} = \frac{1}{3} \times 2.7 \text{ mol per dm}^3 \text{ per second} = 0.9 \text{ mol per dm}^3 \text{ per second}$

[2 marks]

\subsection*{(ii) Rates of formation/consumption}

\[

$$-\frac{d[A]}{dt} = v = 0.9 \text{ mol dm}^{-3} \text{ per second}$$

\]

\[

$$\frac{d[C]}{dt} = v = 0.9 \text{ mol dm}^{-3} \text{ per second}$$

\]

\[

$$\frac{d[D]}{dt} = 2v = 1.8 \text{ mol dm}^{-3} \text{ per second}$$

\]

\textbf{[1 mark each]}

\[

$$\boxed{v = 0.9 \text{ mol dm}^{-3} \text{ per second}, \quad -\frac{d[A]}{dt} = 0.9 \text{ mol dm}^{-3} \text{ per second}, \quad \frac{d[C]}{dt} = 0.9 \text{ mol dm}^{-3} \text{ per second}, \quad \frac{d[D]}{dt} = 1.8 \text{ mol dm}^{-3} \text{ per second}}$$

\]

\section*{Question 3 (5 marks)}

Reaction: $2A \rightarrow P$, second order \\\

Integrated rate law:

\[

$$\frac{1}{[A]_t} - \frac{1}{[A]_0} = 2kt$$

\]

\[

$$k = 4.3 \times 10^{-4} \text{ per molar per second}, \quad [A]_0 = 0.21 \text{ molar}, \quad [A]_t = 0.01 \text{ molar}$$

\]

\[

$$\frac{1}{\text{SI}\{0.01\}\{\text{molar}\}} - \frac{1}{\text{SI}\{0.21\}\{\text{molar}\}} = \text{SI}\{100\}\{\text{per}\{\text{molar}\}} - \text{SI}\{4.7619\}\{\text{per}\{\text{molar}\}} = \text{SI}\{95.2381\}\{\text{per}\{\text{molar}\}}$$

\]

\textbf{[1 mark]}

\[

$$\text{SI}\{95.2381\}\{\text{per}\{\text{molar}\}} = 2 \times (\text{SI}\{4.3\text{e-}4\}\{\text{per}\{\text{molar}\}\{\text{per}\{\text{second}\}}\}) \times t$$

\]

\[

$$\text{SI}\{95.2381\}\{\text{per}\{\text{molar}\}} = \text{SI}\{8.6\text{e-}4\}\{\text{per}\{\text{molar}\}\{\text{per}\{\text{second}\}}\} \times t$$

\]

\[

$$t = \frac{\text{SI}\{95.2381\}\{\text{per}\{\text{molar}\}}}{\text{SI}\{8.6\text{e-}4\}\{\text{per}\{\text{molar}\}\{\text{per}\{\text{second}\}}\}} = \text{SI}\{110742\}\{\text{second}\}$$

\]

\textbf{[2 marks]}

Conversions:

\[

$$t = \frac{\text{SI}\{110742\}\{\text{second}\}}{60} = \text{SI}\{1845.7\}\{\text{minute}\}$$

\]

\[

$$t = \frac{\text{SI}\{1845.7\}\{\text{minute}\}}{60} = \text{SI}\{30.76\}\{\text{hour}\}$$

\]

\textbf{[2 marks]}

$$\boxed{t = 1.107 \times 10^5 \text{ s} = 1.846 \times 10^3 \text{ min} = 30.8 \text{ h}}$$

Question 4 (5 marks)

Arrhenius equation:

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$k_1 = 3.98 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}, \quad T_1 = 288 \text{ K}$$

$$k_2 = 14.3 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}, \quad T_2 = 299 \text{ K}$$

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$\ln \left(\frac{14.3 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}}{3.98 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}} \right) = \ln(3.593) = 1.279$$

[1 mark]

$$\frac{1}{288 \text{ K}} - \frac{1}{299 \text{ K}} = 0.0034722 \text{ K}^{-1} - 0.0033445 \text{ K}^{-1} = 0.0001277 \text{ K}^{-1}$$

\textbf{[1 mark]}

$\left[$

$$1.279 = \frac{E_a}{\text{SI}\{8.314\}\text{J}\text{per}\text{mol}\text{per}\text{K}} \times \text{SI}\{0.0001277\}\text{per}\text{K}$$

$\right]$

$\left[$

$$E_a = \frac{1.279 \times \text{SI}\{8.314\}\text{J}\text{per}\text{mol}\text{per}\text{K}}{\frac{\text{SI}\{10.633\}\text{J}\text{per}\text{mol}}{\text{SI}\{0.0001277\}\text{per}\text{K}}} = \text{SI}\{83260\}\text{J}\text{per}\text{mol}$$

$\right]$

\textbf{[2 marks]}

$\left[$

$$\boxed{E_a = \text{SI}\{83.3\}\text{kJ}\text{per}\text{mol}}$$

$\right]$

\textbf{[1 mark]}

\section*{Question 5 (10 marks)}

Reaction: $A \rightarrow B$

Data:

$\left[$

$\begin{array}{c|cccccc}$

$t/\text{min} \quad 0 \quad 10 \quad 20 \quad 30 \quad 40 \quad \infty$

$\hline$

$[B]/\text{mol} \quad 0 \quad 0.089 \quad 0.153 \quad 0.200 \quad 0.230 \quad 0.312$

$\end{array}$

$\right]$

\subsection*{Step 1: Initial $[A]$ }

$$[A]_0 = 2 \times [B]_{\infty} = 2 \times \text{SI}\{0.312\}\{\text{molar}\} = \text{SI}\{0.624\}\{\text{molar}\}$$

$$[A]_t = [A]_0 - 2[B]_t$$

$$t/\text{min} \quad [B]/\text{mol} \quad [A]/\text{mol}$$

0	0	0.624
10	0.089	0.446
20	0.153	0.318
30	0.200	0.224
40	0.230	0.164

$$t/\text{min} \quad 1/[A]/\text{mol}^{-1}$$

0	1.602
---	-------

$$t/\text{min} \quad 1/[A]/\text{mol}^{-1}$$

0	1.602
---	-------

$$t/\text{min} \quad 1/[A]/\text{mol}^{-1}$$

0	1.602
---	-------

10 & 2.242 \\

20 & 3.145 \\

30 & 4.464 \\

40 & 6.098 \\

\end{array}

\]

\textbf{[2 marks]}

\subsection*{Step 4: Slope and \$k\$}

\[

\text{slope} = \frac{\text{SI}\{6.098\}\{\text{per}\text{molar}\} - \text{SI}\{1.602\}\{\text{per}\text{molar}\}\{\text{SI}\{40\}\{\text{min}\} - \text{SI}\{0\}\{\text{min}\}}}{\text{SI}\{4.496\}\{\text{per}\text{molar}\}\{\text{SI}\{40\}\{\text{min}\}} = \text{SI}\{0.1124\}\{\text{per}\text{molar}\text{per}\text{minute}\}}

\]

For $2A \rightarrow P$: $\frac{1}{[A]_t} - \frac{1}{[A]_0} = 2kt$, so slope $= 2k$:

\[

$k = \frac{\text{SI}\{0.1124\}\{\text{per}\text{molar}\text{per}\text{minute}\}}{2} = \text{SI}\{0.0562\}\{\text{per}\text{molar}\text{per}\text{minute}\}$

\]

\textbf{[3 marks]}

\subsection*{Step 5: Order}

Linear $1/[A]$ vs $t \rightarrow \text{second order}$

\textbf{[1 mark]}

\[

\boxed{\text{Order} = 2, \quad k = \text{SI}\{5.62\text{e-}2\}\{\text{per}\text{molar}\text{per}\text{minute}\}}

\]

\section*{Question 6 (10 marks)}

Electrode: Pt|H₂|H⁺+, \$T = \text{SI}\{298\}\text{K}\$ \\\

Data:

\[

\begin{array}{c|ccccc}

\eta/\text{si}\{\text{milli}\}\text{volt} & 50 & 100 & 150 & 200 & 250 \\\

\hline

j/\text{si}\{\text{milli}\}\text{ampere}\text{per}\text{square}\text{centi}\text{metre} & 2.66 & 8.91 & 29.9 & 100 & 335 \\\

\end{array}

\]

\subsection*{(i) Transfer coefficient \$\alpha\$}

Tafel equation: \$\eta = a + b \ln j\$, \$b = \frac{RT}{\alpha n F}\$ \\\

\$n = 1\$, \$R = \text{SI}\{8.314\}\text{J}\text{per}\text{mol}\text{per}\text{K}\$, \$F = \text{SI}\{96485\}\text{C}\text{per}\text{mol}\$ \\\

\$b\$ in mV:

\[

\$b = \frac{\text{SI}\{8.314\}\text{J}\text{per}\text{mol}\text{per}\text{K} \times \text{SI}\{298\}\text{K}}{\alpha \times 1 \times \text{SI}\{96485\}\text{C}\text{per}\text{mol}} \times 1000 = \frac{\text{SI}\{25.7\}\text{mV}}{\alpha}\$

\]

Compute \$\ln j\$:

\[

\begin{array}{c|c|c}

j/\text{si}\{\text{mA}\text{per}\text{cm}^2\} & \ln(j/\text{si}\{\text{mA}\text{per}\text{cm}^2\}) & \eta/\text{si}\{\text{mV}\} \\\

\hline

2.66 & 0.978 & 50 \\\

8.91 & 2.187 & 100 \\\

29.9 & 3.398 & 150 \\\

100 & 4.605 & 200 \\\

335 & 5.814 & 250 \\

\end{array}

\]

Slope:

\[

$$\frac{\text{SI}\{250\}\{\text{mV}\} - \text{SI}\{50\}\{\text{mV}\}}{5.814 - 0.978} = \frac{\text{SI}\{200\}\{\text{mV}\}}{\text{SI}\{41.36\}\{\text{mV}\}} =$$

\]

Thus:

\[

$$\frac{\text{SI}\{25.7\}\{\text{mV}\}}{\alpha} = \text{SI}\{41.36\}\{\text{mV}\} \quad \rightarrow \quad \alpha = \frac{25.7}{41.36} = 0.62$$

\]

\textbf{[3 marks]}

\textbf{[Comment:]} $\alpha \approx 0.62$ is typical for hydrogen evolution reaction, indicating an asymmetric energy barrier. \textbf{[2 marks]}

\subsection*{(ii) Exchange current density j_0 }

From Tafel:

\[

$$\eta = \frac{RT}{\alpha F} \ln \frac{j}{j_0}$$

\]

Using $\eta = \text{SI}\{50\}\{\text{mV}\} = \text{SI}\{0.050\}\{\text{V}\}$, $j = \text{SI}\{2.66\}\{\text{mA}\per\text{cm}^2\}$:

\[

$$\text{SI}\{0.050\}\{\text{V}\} = \frac{\text{SI}\{0.0257\}\{\text{V}\}}{0.62} \ln \left(\frac{\text{SI}\{2.66\}\{\text{mA}\per\text{cm}^2\}}{j_0} \right)$$

\]

$$\ln\left(\frac{I\{2.66\text{mA}\per\text{cm}^2\}j_0}{I\{0.050\text{V}\}}\right) = \ln\left(\frac{I\{2.66\text{mA}\per\text{cm}^2\}j_0}{I\{0.04145\text{V}\}}\right)$$

$$\ln\left(\frac{I\{2.66\text{mA}\per\text{cm}^2\}j_0}{I\{0.04145\text{V}\}}\right) = \frac{I\{0.050\text{V}\}}{I\{0.04145\text{V}\}} = 1.206$$

$$\frac{I\{2.66\text{mA}\per\text{cm}^2\}j_0}{I\{0.04145\text{V}\}} = e^{1.206} = 3.34$$

$$j_0 = \frac{I\{2.66\text{mA}\per\text{cm}^2\}3.34}{I\{0.796\text{mA}\per\text{cm}^2\}}$$

[3 marks for calculation, 2 marks for final]

$$\boxed{\alpha = 0.62, \quad j_0 = I\{0.80\text{mA}\per\text{cm}^2\}}$$

Question 7 (10 marks)

First-order decomposition in aqueous medium \

Data:

T/K	k/s^{-1}
0	2.46
20	43.5
40	576
60	5480

\]

\subsection*{Step 1: Convert to Kelvin and to \si{\per\second}}

\[

\begin{array}{c|c|c|c|c}

T/K & $1/T/\text{SI}\{e-3\}\text{K}^{-1}$ & $k/\text{SI}\{\text{min}^{-1}\}$ & $k/\text{SI}\{\text{s}^{-1}\}$ & $\ln(k/\text{SI}\{\text{s}^{-1}\})$ \\

\hline

273 & 3.663 & 2.46×10^{-5} & 4.10×10^{-7} & -14.71 \\

293 & 3.413 & 4.35×10^{-4} & 7.25×10^{-6} & -11.83 \\

313 & 3.195 & 5.76×10^{-3} & 9.60×10^{-5} & -9.25 \\

333 & 3.003 & 5.48×10^{-2} & 9.13×10^{-4} & -7.00 \\

\end{array}

\]

\textbf{[3 marks]}

\subsection*{Step 2: Slope from $\ln k$ vs $1/T$ (using $T = 273 \text{ K}$ and 333 K)}

\[

$\text{slope} = \frac{-7.00 - (-14.71)}{\frac{1}{3.003 \times 10^{-3} \text{ K}^{-1}} - \frac{1}{3.663 \times 10^{-3} \text{ K}^{-1}}} =$
 $\frac{7.71}{-0.660 \times 10^{-3} \text{ K}^{-1}} = \text{SI}\{-11681\}\text{K}$

\]

\textbf{[2 marks]}

\subsection*{Step 3: Activation energy}

\[

$E_a = -\text{slope} \times R = \text{SI}\{11681\}\text{K} \times \text{SI}\{8.314\}\text{J}\text{per}\text{mol}\text{per}\text{K} =$
 $\text{SI}\{97100\}\text{J}\text{per}\text{mol} = \text{SI}\{97.1\}\text{kJ}\text{per}\text{mol}$

\]

\textbf{[2 marks]}

\subsection*{Step 4: Pre-exponential factor \$A\$}

Using data at \$T = \text{SI}\{273\}\text{K}\$:

\[

$$\ln k = \ln A - \frac{E_a}{RT}$$

\]

\[

$$-14.71 = \ln A - \frac{\text{SI}\{97100\}\text{J}\text{per}\text{mol}}{\text{SI}\{8.314\}\text{J}\text{per}\text{mol}\text{per}\text{K}} \times \text{SI}\{273\}\text{K}$$

\]

\[

$$-14.71 = \ln A - \frac{\text{SI}\{97100\}\text{J}\text{per}\text{mol}}{\text{SI}\{2270\}\text{J}\text{per}\text{mol}}$$

\]

\[

$$-14.71 = \ln A - 42.78$$

\]

\[

$$\ln A = 28.07$$

\]

\[

$$A = e^{28.07} = \text{SI}\{1.55\text{e}12\}\text{per}\text{second}$$

\]

\textbf{[3 marks]}

\[

$$\boxed{E_a = \text{SI}\{97.1\}\text{kJ}\text{per}\text{mol}, \quad A = \text{SI}\{1.55\text{e}12\}\text{per}\text{second}}$$

\]

\section*{Graph Notes for Students}

`\begin{itemize}`

`\item \textbf{Question 5:}` Plot $1/[A]$ (y-axis) vs t (x-axis). A straight line confirms second order. Slope $= 2k$.

`\item \textbf{Question 6:}` Plot η (mV) on y-axis vs $\ln j$ on x-axis. Slope $= b = RT/(\alpha F)$, intercept gives j_0 .

`\end{itemize}`

`\end{document}`